Antireflux Surgery and Its Alternatives. Nissen versus Partial Wraps, Recurrence, Jejunal Feeding and Outcomes in Children with Neurological Impairment.

Picture a seven-year-old boy with severe cerebral palsy. He cannot sit without support, he is fed through a gastrostomy, and this year he has been admitted twice with aspiration pneumonia despite a proton pump inhibitor. At the multidisciplinary meeting the surgeon offers a laparoscopic Nissen fundoplication. The interventional radiologist points out that a gastrojejunal tube could be placed instead, without an operation. His parents want to know which is safer, which will keep him out of hospital, and whether the operation will last. This chapter is about how to answer them.

The main idea is the following. Antireflux surgery in children is reserved for a small group: those with life-threatening complications of reflux, those whose symptoms are refractory to properly optimised medical treatment, and those with chronic conditions that carry a high risk of harm from reflux. In the only paediatric randomised trial comparing wraps, a full Nissen wrap failed less often than a partial wrap but caused more severe swallowing problems, and almost all of the failures happened in neurologically impaired children. Over a decade, laparoscopic wraps recur more often than open ones. For the neurologically impaired child, feeding directly into the jejunum is a legitimate alternative to a wrap, and no randomised trial tells us which route is better. The choice is therefore made with the family, using honest numbers.

A few terms first. Gastro-oesophageal reflux disease, or GERD, means reflux of stomach contents that causes troublesome symptoms or complications. Fundoplication is an operation in which the upper part of the stomach, the fundus, is wrapped around the lower oesophagus to strengthen the barrier against reflux. A Nissen fundoplication is a complete wrap of three hundred and sixty degrees. Partial wraps go only part of the way round: the Toupet wrap covers about two hundred and seventy degrees from behind, the Thal wrap is a partial wrap from the front, and the Dor wrap covers about one hundred and eighty degrees from the front. A gastrojejunal tube, often shortened to GJ tube, enters the stomach through a gastrostomy, but its tip is passed through the pylorus into the jejunum, so that feeds bypass the stomach. Total oesophagogastric dissociation is a much larger operation in which the oesophagus is disconnected from the stomach and joined to a loop of jejunum, while the stomach is fed through a gastrostomy. Neurological impairment, sometimes shortened to NI, describes children with conditions such as severe cerebral palsy. The Gross Motor Function Classification System, or GMFCS, grades motor disability from level one to level five, the most severe.

The sources for this chapter include the 2018 paediatric reflux guideline of the North American and European societies for paediatric gastroenterology, hepatology and nutrition, written by Rosen and colleagues; the 2017 European guideline on gastrointestinal complications in neurologically impaired children by Romano and colleagues; the 2019 European position paper on jejunal feeding; the adult guidelines of the American College of Gastroenterology from 2022, the American Gastroenterological Association practice update from 2022, and the 2023 multi-society guideline led by the Society of American Gastrointestinal and Endoscopic Surgeons; the Oxford randomised trial of Nissen versus Thal by Kubiak and colleagues; an adult meta-analysis of Nissen versus Toupet by Broeders and colleagues; the Oslo randomised trial of laparoscopic versus open surgery with twelve years of

follow-up by Fyhn and colleagues; a London trial of the same comparison; a Kansas City trial of oesophageal mobilisation by St Peter and colleagues; studies of gastrojejunal feeding and of quality of life; and a Cochrane review published in 2026.

Part one. What a wrap does, and what it costs.

A fundoplication works through several mechanisms at once. It raises the resting pressure at the lower oesophageal sphincter. It restores a segment of oesophagus inside the abdomen, where the positive abdominal pressure helps keep it closed. It reduces the number of transient relaxations of the lower oesophageal sphincter, which are the main way reflux happens in both children and adults. And the surgeon closes the crura of the diaphragm around the oesophagus, which repairs any hiatus hernia and recreates the sharp angle at which the oesophagus enters the stomach.

Each of these benefits has a matching cost. A wrap that stops stomach contents coming up also makes it harder for gas to come up, so the child may be unable to belch and may develop gas bloat. It makes it harder to vomit, which matters in a child who retches. A tight or long wrap can cause dysphagia, meaning difficulty swallowing. And the operation can speed up gastric emptying, which can cause dumping syndrome.

It is just as important to understand what a wrap cannot do. A wrap stops stomach contents from rising. It does not stop a child from aspirating his own saliva, it does not fix poor coordination of swallowing in the mouth and throat, and it does not treat gastric dysmotility or retching that comes from the brain rather than the stomach. In neurologically impaired children, much of the respiratory harm comes from above rather than from the stomach. That is why outcomes after a technically perfect wrap can still disappoint.

Part two. Who should be considered for surgery.

The 2018 paediatric guideline says that antireflux surgery should only be considered in children with GERD who have one of four things: life-threatening complications; symptoms refractory to optimal therapy, after appropriate evaluation to exclude other underlying diseases; chronic conditions, such as neurological impairment or cystic fibrosis, with a significant risk of reflux-related complications; or the need for chronic drug treatment to control symptoms. The same guideline lists feeding into the small bowel, past the pylorus, as an alternative to fundoplication for refractory reflux. It also states, with moderate evidence, that a trial of jejunal feeding should not be used to diagnose GERD.

The 2017 European guideline for neurologically impaired children is more specific. It recommends strongly, based on high-quality evidence, that a routine antireflux procedure should not be performed at the time a gastrostomy is placed. It recommends that fundoplication be considered when optimised medical therapy has failed. It suggests jejunal feeding when aspiration is caused by reflux, although that suggestion is weak. And it suggests that total oesophagogastric dissociation be restricted to very selected cases.

Before booking the operating theatre, several steps prevent an unnecessary or failing wrap. First, prove that reflux is the problem with objective testing, such as pH-impedance monitoring or endoscopy, rather than relying on symptoms. Second, exclude conditions that mimic reflux: eosinophilic oesophagitis, rumination, constipation, cow's milk protein allergy and gastroparesis. Third, check the

anatomy with a contrast study, looking for malrotation, a hiatus hernia or obstruction at the outlet of the stomach. Fourth, ask whether aspiration is coming from above; a videofluoroscopic swallow study can change the whole goal of treatment. And fifth, take retching and delayed gastric emptying seriously, because they predict a poor result, and in such a child jejunal feeding may be the better first step. A trial of jejunal feeding cannot prove reflux, but it can show the family how their child behaves when nothing enters the stomach, and some families decide to stay with it.

Part three. Which wrap.

The only randomised trial of wrap type in children was performed in Oxford by Kubiak and colleagues and published in the Annals of Surgery in 2011. One hundred and seventy-five children were randomised to a Nissen or a Thal fundoplication by a single surgeon, and one hundred and sixty-seven were analysed after a median follow-up of thirty months. The main outcome was absolute failure, defined as needing a redo fundoplication or a feeding jejunostomy through the stomach. This happened in about six percent of children after a Nissen and about sixteen percent after a Thal, a significant difference. Redo fundoplication was needed in about five percent after a Nissen and about fifteen percent after a Thal. On the other hand, severe dysphagia, needing endoscopy or dilatation, occurred in about twelve percent after a Nissen and only about two percent after a Thal. Any degree of dysphagia was similar, at about twenty-two to twenty-four percent in both groups, and the proportion who restarted antireflux medicine was also similar. The most striking finding was where the failures happened. Seventeen of the eighteen absolute failures occurred in neurologically impaired children. Among neurologically normal children, no Nissen and only one Thal failed. In other words, in neurologically normal children either wrap usually works, and in impaired children the Nissen is more durable, at the price of more dilatations.

Adult data point the other way. A meta-analysis by Broeders and colleagues in the British Journal of Surgery in 2010 pooled seven adult randomised trials comparing a Nissen with a Toupet wrap in about eight hundred patients. Reflux recurred equally often after both. But after a Nissen, patients were about one and a half times as likely to have dysphagia and gas bloat, about twice as likely to be unable to belch, about two and a half times as likely to need dilatation, and about twice as likely to need reoperation. For this reason, the 2023 adult multi-society guideline suggests a partial wrap over a complete one.

In children, the pooled evidence is too thin to settle the matter. A meta-analysis by Glen and colleagues in 2014 found only two paediatric randomised trials and no significant difference in success, with an odds ratio of about one point three and a wide confidence interval; in the neurologically impaired subgroup the confidence interval was wider still. A systematic review of prospective paediatric series by Mauritz and colleagues in 2011 found a median success rate of eighty-six percent, similar recurrence after total and partial wraps, and less dysphagia after partial wraps. So the honest summary is this: in adults, the Toupet controls reflux as well as the Nissen with fewer side effects; in children, the one trial favours the Nissen for durability in impaired children, and the choice of wrap is still a matter of surgical judgement.

Part four. Laparoscopic or open, and how the wrap is built.

Most fundoplications in children are now laparoscopic, because recovery is faster and the hospital stay shorter. But the long-term data are sobering. In the Oslo randomised trial, eighty-seven children were assigned to laparoscopic or open fundoplication. After a median of four years, reflux had recurred in thirty-seven percent after laparoscopy and seven percent after open surgery, a risk about five times higher. The same children were followed for a median of about twelve years. By then reflux had recurred in fifty-six percent after laparoscopy and thirty-one percent after open surgery, a hazard ratio of about two and a half. Recurrence also came earlier after laparoscopy, at a median of one year compared with five years. Redo surgery was needed in nineteen percent versus seven percent, a difference that did not reach significance. Reassuringly, at the end of follow-up most patients had no reflux symptoms and more than four in five were fully satisfied. A smaller London trial, mostly of neurologically impaired children, found recurrence of twenty percent after laparoscopy and about thirteen percent after open surgery at four years, not a significant difference, but retching was far less common after laparoscopy, seven percent compared with fifty percent. Laparoscopy is still the standard approach, but families should hear that a wrap is not a lifetime guarantee.

How the wrap is built also matters. In a randomised trial of one hundred and seventy-seven children by St Peter and colleagues, minimal dissection of the oesophagus, which leaves its attachments to the diaphragm in place, was compared with extensive mobilisation. Herniation of the wrap into the chest occurred in about eight percent after minimal dissection and thirty percent after extensive dissection, and reoperation was needed in about three percent versus eighteen percent. In a single surgeon's series of more than one thousand laparoscopic Nissen operations, wrap failure occurred in four percent, postoperative complications in four percent, and the average hospital stay was just over one day. But a critical review of thirty-six paediatric studies by Martin and colleagues showed how much published results vary: recurrence ranged from zero to forty-eight percent, reoperation from under one percent to almost eighteen percent, and mortality from zero to twenty-four percent, the last reflecting how sick the operated populations were. Only about one study in six actually defined what it meant by recurrence. So before quoting a number to a family, ask how recurrence was measured.

Families should also hear about the side effects. Severe dysphagia occurred in about twelve percent after a Nissen in the Oxford trial; most settles, but some children need dilatation. Gas bloat and inability to vomit are common; a venting gastrostomy helps. Retching can persist or even appear for the first time, because in impaired children it is often central rather than caused by reflux. Dumping syndrome is easily missed. In one series of one hundred and ninety children it occurred in nine percent, and the risk was much higher when surgery was done before twelve months of age, in children with severe scoliosis, and in children with a very small stomach. Because the wrap speeds up gastric emptying, a child may become pale, sweaty, lethargic or even have seizures an hour or two after a feed, from reactive hypoglycaemia. Check the blood glucose after a feed in any child whose behaviour changes after a wrap. Finally, redo surgery is harder: in a series of redo fundoplications, a quarter of children had early major complications and about a fifth had recurrence again.

Part five. Jejunal feeding as the alternative.

For a neurologically impaired child who already has a gastrostomy, the main alternative to fundoplication is to convert the gastrostomy to a gastrojejunal tube. Feeds then go directly into the jejunum and the stomach is bypassed. Reflux is not abolished, because the stomach still produces acid

and saliva still collects, but the volume available to reflux is much smaller.

How do the two strategies compare? A Cochrane review published in 2026 searched for all comparative studies in neurologically impaired children. It found no randomised trials at all, only eleven non-randomised studies including about three thousand one hundred children. For admissions with pneumonia, there was little or no difference between gastrostomy with fundoplication and gastrostomy alone, and little or no difference between fundoplication and a gastrojejunal tube; the certainty of this evidence was low or very low. A matched study from United States children's hospitals compared one hundred and fourteen pairs of children and found about two reflux-related admissions per child per year after a gastrojejunal tube and about one point seven after fundoplication, not a significant difference. A cohort of three hundred and sixty-six children from Ontario, reported by Srivastava and colleagues in Pediatrics in 2009, found hazard ratios of about zero point seven for aspiration pneumonia and about zero point five for death after fundoplication compared with a gastrojejunal tube, but neither difference was statistically significant. The overall picture is that neither route clearly prevents more pneumonia or death.

Gastrojejunal tubes avoid an operation but not risk. A systematic review of more than two thousand seven hundred procedures found bowel perforation in about two percent overall: about three percent in children under ten kilograms, compared with about one in a thousand in larger children. Most perforations occurred in the first month. Mortality was just under one percent per patient. Tubes also dislodge and block. In one series, low-profile gastrojejunal tubes lasted on average about three months before needing replacement, compared with about six months for a separate gastrostomy and jejunal extension. In a small Utah series, tube-related visits rose more than five-fold after a gastrojejunal tube was placed, and about one in five children later had a fundoplication anyway. Feeds must usually run continuously, which ties the child to a pump for many hours. The 2019 European position paper recommends checking copper, zinc, selenium and iron every six to twelve months in children on long-term jejunal feeds.

The third option, total oesophagogastric dissociation, is reserved for very selected children, usually those with the most severe impairment or a failed redo wrap. In a British comparison of children at GMFCS level five, by Lansdale and colleagues in 2015, symptoms recurred in one of twenty-three after dissociation compared with five of twenty-four after laparoscopic fundoplication, which was not a significant difference, and only four children after dissociation still needed acid suppression compared with thirteen after fundoplication. Carers rated quality of life similarly. Dissociation abolishes reflux almost completely, but it is a major, irreversible operation.

Putting it together, the three options can be compared feature by feature. Gastrostomy with fundoplication is one operation, allows bolus feeds into the stomach, controls reflux well but may fail over the years, and carries risks of dysphagia, bloat and dumping. A gastrojejunal tube is placed by a radiologist and replaced repeatedly, requires continuous feeds, bypasses rather than abolishes reflux, carries risks of dislodgement, perforation and intussusception, and is fully reversible. Dissociation is a major operation that abolishes reflux, is irreversible and is kept for very selected cases. Present these options side by side to the family, with the numbers, and record why the route was chosen.

Part six. What families actually gain, and how practice is changing.

What do children with severe impairment and their carers gain from a wrap? In a small prospective study by Srivastava and colleagues in 2007, children's bodily pain scores and their physical role limitation scores improved one month after fundoplication, and carers' vitality improved. But parenting stress did not change and remained above the population norm. And as the Oxford trial showed, wrap failure concentrates in exactly the children who most need the operation to work.

Practice has changed. A national study of almost forty-nine thousand antireflux procedures in the United States between 1996 and 2003 by Lasser and colleagues found that the proportion performed in neurologically impaired children fell from fifty-three to forty percent. A study of more than thirteen thousand procedures in thirty-six children's hospitals by Goldin and colleagues in 2009 found that rates were falling every year and varied widely between hospitals. And a more recent analysis of children's hospital data found that the average number of fundoplications per institution fell from about fifty to about seventeen a year between 2010 and 2019, while the number logged by surgical trainees fell from forty-six to twenty-six. Variation larger than the evidence can explain means that the operation a child receives depends partly on where he lives, and falling volume means that fewer surgeons have the experience that the technique trials show to matter.

Part seven. Newer options and what the guidelines say.

Adolescents and parents sometimes ask about the newer adult techniques. Transoral incisionless fundoplication, often shortened to TIF, creates a partial wrap through an endoscope. In a small American series of eleven children, mostly neurologically impaired, reflux resolved in ten at about eight months. Magnetic sphincter augmentation, sold as LINX, places a ring of magnetic beads around the lower oesophagus; the pivotal adult trial enrolled only people aged twenty-one and older, and there is no published paediatric series. Radiofrequency treatment of the sphincter, known as Stretta, has no paediatric data. In children all three remain experimental. Adult guidelines from the American Gastroenterological Association and the 2023 multi-society guideline accept magnetic augmentation and TIF for carefully selected adults without a large hiatus hernia, but paediatric guidelines do not mention them. An adolescent who asks for one should be referred to a programme with adult experience and an outcome registry.

When the paediatric and adult guidelines are compared, they agree on who should be considered for surgery: patients with objectively proven reflux that is refractory, complicated or dependent on long-term medication. They differ on wrap type, because the adult guideline prefers partial wraps while paediatric guidelines say nothing. There are gaps in both directions. Only the paediatric documents address routine wraps at gastrostomy, jejunal feeding and dissociation, and only the adult documents address endoscopic and device therapy, which is the most material difference. In short, adults are offered partial wraps and devices that have no paediatric evidence, and children are offered jejunal feeding that adult guidelines never mention.

Part eight. Traps and pearls.

There are several common traps. The first is to add a wrap to every gastrostomy just in case; the European guideline strongly advises against this, and most children never need one. The second is to promise that a wrap will stop the pneumonias; saliva is still aspirated, and no study shows a clear fall in pneumonia. The third is to treat a jejunal feeding trial as proof of reflux; it is not a diagnostic test, so

use pH-impedance. The fourth is to assume that symptoms after a wrap cannot be reflux; recurrence reaches thirty to fifty-six percent by twelve years, so re-test. The fifth is to label pallor, sweating or seizures after feeds as neurological events without checking glucose; think of dumping. And a child who retches after a wrap has not necessarily been given a wrap that is too tight; in impaired children retching is often central and was there before the operation.

The numbers worth carrying into clinic are these. Surgery is for life-threatening complications, refractory disease, high-risk chronic conditions and long-term drug dependence. There should be no routine wrap at gastrostomy in neurologically impaired children. In the Oxford trial, the Nissen failed in about six percent and the Thal in about sixteen percent, severe dysphagia occurred in about twelve versus two percent, and seventeen of the eighteen failures were in impaired children. In adults, a Nissen roughly doubles reoperation and dilatation compared with a Toupet, with equal control of reflux. In the Oslo trial, reflux recurred in fifty-six percent after laparoscopic and thirty-one percent after open surgery by twelve years. Minimal mobilisation cut wrap herniation from thirty to about eight percent and reoperation from about eighteen to three percent. Dumping occurs in about nine percent, especially after surgery in infancy. Wrap and gastrojejunal tube have never been compared in a randomised trial, and pneumonia and mortality are not clearly different. Gastrojejunal tubes perforate in about three percent of children under ten kilograms and last about three months. And dissociation almost abolishes reflux in the most severely impaired children, but it is irreversible.

What should you do differently in clinic? Do not refer a child for a wrap until reflux has been proven objectively and its mimics excluded. Ask whether aspiration comes from above, and look for retching, delayed emptying and dumping risk before surgery. When a neurologically impaired child needs a gastrostomy, place it without a wrap unless there is a clear indication. When a wrap is being considered, set out the alternatives side by side, including a gastrojejunal tube, with the numbers on failure, recurrence, perforation and tube changes, and let the family choose what they can live with. After a wrap, check glucose when a child changes after feeds, and re-test rather than assume when symptoms return. For the boy at the start of this chapter, that means a swallow study, an honest conversation about wrap versus tube, and a shared decision rather than a default.